

HANDLEGUARD AI

A Privacy-Preserving Temporal Video Intelligence Platform for Damage-Free Warehouse Handling

Complete Solution Proposal

Prepared in response to the “AI Video Intelligence for Warehouse Handling” challenge
September 2026

Operational user journey



The assistant never treats “risk” as “confirmed damage”; significant incidents remain reviewable by a human supervisor.

Proposed value proposition: turn existing cameras into an event-aware operational assistant that identifies risky handling sequences early, explains the evidence, and supports corrective action without identity-based worker surveillance.

Pilot scope 1 loading bay / 1 process	Behaviour target 10 scenarios	Design priority Prevention > retrospective review
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Executive Decision Summary

Assessment of the supplied challenge brief. The brief is strong as a requirements document, but it is not yet a complete proposal. It specifies the problem, expected capabilities, behaviours, responsible-AI principles, deliverables, and judging criteria; however, it does not define a unique solution architecture, implementable detection logic, data strategy, risk model, evaluation protocol, deployment design, business-impact methodology, or execution plan. This proposal fills those gaps while staying within the challenge scope.

Proposed solution	HandleGuard AI - an event-centric video intelligence platform that combines detection, tracking, pose/scene geometry, temporal reasoning, risk scoring, evidence storage, and a grounded supervisor assistant.
Primary novelty	A temporal Event Graph and hybrid risk engine reason over person-product-pallet-equipment interactions across time instead of classifying isolated frames.
Safe AI posture	No face recognition, no identity ranking, no automatic punitive action. Alerts carry evidence, uncertainty, and require human review for consequential decisions.
Prototype scope	One controlled loading/unloading process, limited camera angles, 10 predefined scenarios, real-time or near-real-time event replay and dashboard.
Success definition	High event-level precision/recall, low nuisance-alert rate, usable explanations, low alert latency, and measurable reduction in repeated risky handling during the pilot.

What was enhanced

- Converted the challenge statement into a specific product concept with a defensible technical architecture and clear non-goals.
- Resolved the brief's scope ambiguity by targeting 10 scenarios: the brief first suggests 4/5 behaviours but later requires the working prototype to demonstrate at least 10 predefined behaviours/scenarios.
- Added behaviour-by-behaviour detection logic, temporal evidence requirements, and a calibrated risk score rather than a generic "AI detects bad handling" claim.
- Added an evidence-bound LLM assistant that answers only from detected incidents, SOPs, and stored event metadata, preventing fabricated incident narratives.
- Added privacy-by-design controls, human review gates, data minimization, retention, auditability, and non-identifying analytics.
- Added an evaluation plan aligned to the judging weights: technical execution, AI/video integration, damage prevention, UX, innovation, and presentation quality.
- Added a six-day submission sprint and a post-submission pilot plan so the proposal remains achievable rather than becoming an over-scoped research prototype.

1. Challenge Understanding and Opportunity

Warehouses already have cameras, but conventional CCTV creates evidence after an incident rather than prevention before damage. The target system should therefore perform a sequence of capabilities: perceive entities, understand how they interact over time, infer handling risk, preserve evidence, alert a supervisor, and learn from review outcomes. The core outcome is operational intelligence, not people monitoring.

Operational user journey



The assistant never treats “risk” as “confirmed damage”; significant incidents remain reviewable by a human supervisor.

Figure 1. From passive recording to proactive operational intelligence.

1.1 Primary users

User	Decision supported
Warehouse supervisor	Needs high-confidence incidents, replay, trends, and fast explanation.
Loading/unloading operator	Needs timely, non-punitive guidance and clear safe-handling cues.
Logistics manager	Needs process-level risk patterns, bay comparison, and preventable-loss indicators.
Quality professional	Needs evidence linking handling events to inspection or rework decisions.
Safety professional	Needs repeatable behaviour taxonomy, audit trail, and human-review workflow.

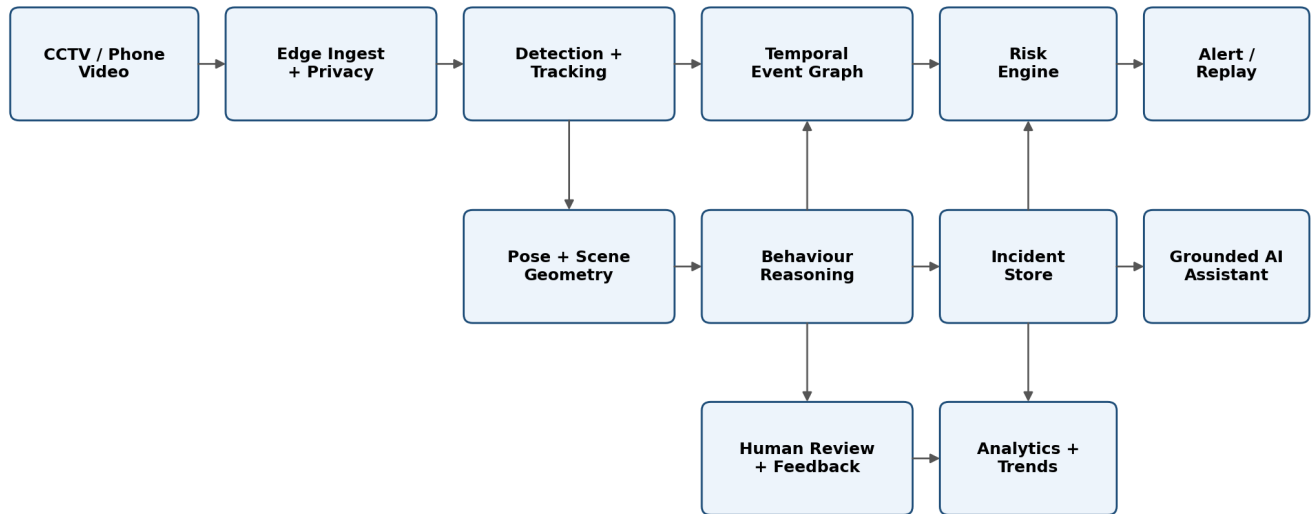
1.2 Design goals and non-goals

- Goal: detect risky handling behaviours as events with start/end timestamps and visual evidence.
- Goal: provide near-real-time alerts only when event confidence and risk justify interruption.
- Goal: distinguish observed behaviour, potential risk, and confirmed damage.
- Goal: work with existing CCTV or smartphone video and support limited-edge deployment.
- Non-goal: biometric identification, employee ranking, or automated disciplinary decisions.
- Non-goal: claim physical damage when only a risky handling event is visible.
- Non-goal: replace human inspection for high-consequence quality or safety decisions.

2. Proposed Solution: HandleGuard AI

HandleGuard AI is a hybrid computer-vision and event-reasoning system. Instead of asking a model to label every frame as “safe” or “unsafe,” it tracks entities and constructs short temporal interaction histories: who/what moved, where it moved, how fast, whether contact occurred, whether support geometry became unstable, and what happened immediately before and after the event.

HandleGuard AI - Event-Centric Warehouse Video Intelligence Architecture



Core principle: detect entities, reason over interactions across time, quantify risk, preserve evidence, and route verified facts to humans.

Figure 2. Proposed modular architecture. The LLM consumes incident evidence; it does not independently invent warehouse events.

2.1 Core modules

- 1. Video ingest and privacy gate:** RTSP/file/mobile video ingestion; optional face blurring at the edge; frame sampling; time synchronization.
- 2. Scene perception:** Detect people, cartons/products, pallets, trolleys, forklifts and configured zones; optional pose keypoints for foot/hand interactions.
- 3. Multi-object tracking:** Maintain stable IDs across frames to calculate trajectories, velocity, proximity, hand-off, and object-person association.
- 4. Temporal Event Graph:** Represent interactions such as person-holds-box, box-above-floor, box-over-pallet, box-enters-zone, contact-released, impact-likely.
- 5. Behaviour reasoning:** Combine learned action classification with deterministic temporal/geometry rules for explainable event decisions.
- 6. Risk engine:** Estimate severity from behaviour type, kinematics, stack/support geometry, duration, recurrence, zone context, and confidence.
- 7. Incident evidence store:** Save timestamps, representative frames, short replay clips, entities, risk components, model confidence, review state, and optional SOP reference.
- 8. Grounded AI supervisor:** Answer questions only from the incident store and approved SOPs; cite timestamps/incident IDs in every factual response.
- 9. Dashboard and feedback:** Timeline, live alerts, replay, trends, heatmaps, filters, acknowledgement, false-positive labeling, and improvement tracking.

3. Innovation and Differentiation

Differentiator	Why it matters
Event-centric reasoning instead of frame-only detection	Damage-causing actions are sequences. The system models transitions such as lift → move → release → rapid descent → impact → stationary product, making the evidence interpretable.
Hybrid learned + rule-based behaviour intelligence	Object/action models handle visual variability; temporal and geometry rules constrain impossible or weak interpretations. This reduces reliance on a single opaque classifier.
Pre-incident and early-risk signals	The system can flag unstable stack geometry, excessive pallet overhang, risky approach trajectories, or repeated mishandling before a damage claim is made.
Evidence-bound conversational AI	Every assistant answer is retrieved from incident records and SOP text. Unsupported questions return “insufficient evidence” rather than a plausible but fabricated answer.
Uncertainty-aware alerting	Low-confidence events are queued for review instead of generating high-severity alarms. Confidence and risk are separate fields.
Privacy-preserving analytics	Worker-independent event metrics, local anonymization, minimal clip retention, role-based access, and no identity scorecards by default.
Closed-loop improvement	Supervisor dispositions - true event, false positive, severity correction, action taken - become structured feedback for threshold tuning and active learning.

4. Behaviour Taxonomy and Detection Logic

To satisfy the strictest reading of the brief, the prototype targets 10 distinct scenarios. Each event requires temporal evidence rather than a single-frame label. Thresholds are configurable by product category and camera geometry.

Scenario	Detection logic	Evidence shown
B1 Drop / throw	Track product release, rapid downward/ballistic motion, likely impact, and post-impact state.	Clip from 1-2 s before release through impact; trajectory and confidence.

B2 Dragging instead of proper movement	Product centroid moves horizontally near floor while associated with a person and lacks trolley/pallet support.	Trajectory, ground-contact proxy, operator-product association.
B3 Rough handling / excessive impact	High relative acceleration or abrupt deceleration around contact/placement; repeated impact cues increase severity.	Motion curve, contact frame, replay.
B4 Incorrect weight/size stacking	Configured heavy/large class above lighter/smaller item, or support hierarchy violates product rules.	Detected stack order and product-class metadata.
B5 Unstable stack / insufficient support	Low overlap/support ratio, large overhang, visible tilt, or center projection near/outside support polygon.	Stack geometry overlay and threshold values.
B6 Product outside designated staging/loading area	Tracked product enters or remains in a forbidden/incorrect polygon beyond allowed duration.	Zone ID, dwell time, entry/exit timestamps.
B7 Improper equipment / manual handling of large product	Large/heavy configured product moved without trolley/pallet truck/team-lift cue.	Product class/size, person count, equipment presence.
B8 Pallet misposition / product overhang	Pallet-product alignment violates configured support margin or pallet is outside docking region.	Pallet/product geometry and zone reference.
B9 Stepping/standing/rolling on product	Foot keypoints overlap package surface for minimum duration, or product rolling pattern violates rules.	Pose-product overlap and persistence.
B10 Unsafe loading/unloading sequence	Finite-state process violation, e.g., unstable staging → move → unsupported placement, or configured step omitted.	Sequence trace and violated SOP state transition.

4.1 Event-state example: dropped carton

1. t0: carton supported by operator
2. t1: carton released
3. t2: vertical velocity increases beyond threshold
4. t3: carton intersects floor/stack boundary

5. t4: abrupt velocity reduction = impact proxy
6. t5: product stationary or orientation changes
7. Decision: “potential high-risk drop event,” not “confirmed damage”
8. Action: replay + inspect-product recommendation + supervisor acknowledgement

5. Risk Scoring and Alert Policy

Risk should not be a black-box label. HandleGuard AI computes an interpretable 0-100 operational risk score from normalized components. A prototype formulation is:

$$\text{Risk} = 100 \times [0.30B + 0.25K + 0.20S + 0.10D + 0.10F + 0.05Z]$$

where B = base severity of behaviour; K = motion/impact intensity; S = stacking/support instability; D = duration; F = repeated-frequency factor; and Z = zone/process context. Components not applicable to an event are re-normalized. Model confidence is reported separately and can gate alerts.

Risk band	System action
0-24 Low	Log for analytics; no interruption unless repeated.
25-49 Medium	Show in dashboard; optional supervisor digest.
50-74 High	Near-real-time alert with replay and recommended check.
75-100 Critical	Immediate alert; require acknowledgement and human review.

5.1 Alert suppression and human review

- Use hysteresis/cooldown so one continuous event does not create dozens of alerts.
- Require minimum event duration or multi-cue agreement for ambiguous behaviours.
- Demote low-confidence high-risk candidates to a review queue rather than a critical alarm.
- Allow supervisors to mark false positive, correct event, wrong severity, duplicate, or uncertain.
- Never use an unreviewed event for punitive personnel action.

6. AI Operations Assistant

The assistant is not the event detector. It is a grounded interface over already-detected evidence and approved operational documents. This separation is essential for reliability.

Supervisor query	Grounded response behaviour
“Show today’s high-risk unloading events.”	Query incident store by date/process/risk; return event IDs, timestamps, thumbnails and replay links.
“Why was event HG-014 high risk?”	Return the stored behaviour, risk components, confidence, evidence frame, and matching SOP rule.

“Which bay repeated dragging most often?”	Aggregate reviewed dragging events by zone; show counts and trend.
“Did this carton get damaged?”	If video only shows risky handling, answer that damage is unconfirmed and recommend inspection.
“Who is the worst worker?”	Refuse identity ranking in default mode; redirect to process/behaviour trends.

7. Data Strategy and Annotation Protocol

7.1 Data sources

- Organizer-provided warehouse videos where authorization permits their use.
- Controlled staged recordings with boxes, pallets/trolleys and volunteers to cover rare behaviours safely.
- Public or synthetic footage only when licensing, domain relevance and labeling quality are verified.
- Hard-negative clips showing normal handling that visually resembles risky handling; these are essential to control false positives.

7.2 Annotation unit

The primary label is an event interval, not a frame. Each annotation stores: video/session ID; start/end time; behaviour; involved entity tracks; zone; severity label; evidence note; visibility/occlusion; camera angle; reviewer; and optional uncertainty. This enables event-level metrics and avoids inflated accuracy from thousands of redundant frames.

7.3 Split strategy

- Split by source video/session, not random frames, to prevent temporal leakage.
- Keep staged scenarios from the same take in one partition.
- Maintain a held-out camera angle or session for robustness testing.
- Report per-behaviour results so dominant classes do not hide weak rare-event performance.

8. Technical Architecture and Stack

Layer	Proposed implementation
Video processing	OpenCV / GStreamer / FFmpeg; RTSP or file ingestion; adaptive frame sampling.
Detection	Lightweight YOLO-family detector or RT-DETR variant; fine-tuned only on warehouse-specific classes when required.
Tracking	ByteTrack-class multi-object tracking with track persistence and occlusion handling.

Pose / geometry	Lightweight pose estimation for hand/foot cues plus calibrated floor/zone polygons and support-overlap calculations.
Temporal reasoning	Short-window feature buffer + finite-state temporal rules; optional temporal action model for ambiguous classes.
Risk engine	Python rule/model service with configurable thresholds, uncertainty and behaviour-specific weighting.
Backend	FastAPI; event API, alert service, assistant retrieval service, audit logging.
Storage	PostgreSQL for production-style event records or SQLite for the competition prototype; local/object storage for short evidence clips.
Frontend	React/Next.js or equivalent responsive web UI for live feed, incident timeline, trends and assistant.
LLM layer	Swappable small instruct model or hosted API; retrieval constrained to event DB + SOP corpus; structured tool calls only for incident facts.
Deployment	Docker; ONNX/TensorRT/OpenVINO where supported; edge-first anonymization and optional cloud analytics.

8.1 Why a hybrid approach is preferable for the prototype

Recent warehouse-safety research already combines object detection, tracking, action recognition and alarm logic. The differentiator should therefore not be “we used YOLO.” HandleGuard AI uses perception as an input to a transparent event model, adds geometry/temporal constraints, exposes evidence and uncertainty, and measures whether alerts actually reduce repeated risky handling. This keeps the prototype feasible while providing a stronger systems contribution than a single classifier.

9. Evaluation and Success Metrics

Targets below are proposal goals, not claimed results. Final reporting should clearly separate measured values from targets.

Dimension	Metric	Interpretation
Object detection	mAP@0.5 and mAP@0.5:0.95 by class	Establish perception quality; no single global accuracy claim.

Tracking	IDF1 / HOTA or ID switches on selected clips	Check whether temporal reasoning can trust trajectories.
Behaviour events	Event precision, recall, macro-F1 by behaviour	Primary AI success measure; use interval/event matching.
False alarms	False positive events per video-hour / shift	Directly reflects supervisor burden.
Latency	p50 and p95 detection-to-alert latency	Demonstrate near-real-time practicality.
Risk calibration	Brier score / reliability plot where graded labels exist	Avoid treating confidence as calibrated by default.
Assistant grounding	Evidence/timestamp correctness on scripted queries	Every factual claim should trace to stored evidence.
Operational	Repeat risky-event rate before/after intervention in staged pilot	Tests prevention, not just detection.
UX	Supervisor task success + 1-5 usefulness score	Measures whether the dashboard actually supports decisions.
Privacy	Identity data stored, retention, access audit	Target: no biometric identity storage in prototype.

9.1 Prototype acceptance targets

- Behaviour macro-F1 target: ≥ 0.80 on a staged/session-separated holdout, with per-class reporting.
- Event precision target: ≥ 0.85 for alert-producing High/Critical events after threshold tuning.
- Event recall target: ≥ 0.80 for the 10 target scenarios in the controlled pilot.
- Alert latency target: $p95 < 2$ seconds on GPU-class hardware or < 5 seconds in a laptop/CPU fallback demonstration.
- Grounded assistant target: 100% of scripted incident answers include a valid incident ID/timestamp or explicitly state insufficient evidence.
- User target: $\geq 4/5$ average usefulness from a small supervisor/safety-style review panel.

10. Pilot Design

10.1 Controlled pilot configuration

Area	One loading/unloading bay or simulated bay.
Process	One repeatable loading/unloading workflow.
Products	2-3 representative carton/package categories.

Equipment	Pallet, trolley/pallet truck proxy, loading table/vehicle boundary.
Cameras	1-2 fixed views plus optional smartphone view.
Scenarios	10 predefined behaviours + normal-handling hard negatives.
Reviewers	At least one technical evaluator and several representative users if available.

10.2 Demonstration script

9. Play a normal unloading sequence: no alert; event store remains clean.
10. Stage a drop: bounding/tracking overlay → temporal event → risk score → alert → 5-10 second incident replay.
11. Stage dragging and unstable stack: show different evidence features and risk components.
12. Ask the assistant for high-risk events and why one event was flagged; it returns incident-backed facts.
13. Mark one alert false positive and show the feedback/audit trail.
14. Open shift summary showing behaviour counts, trend, zones and preventive recommendations.

11. Damage Prevention and Business Impact

The proposal should avoid inventing savings. Instead, the dashboard can compute an auditable avoided-loss estimate from organization-provided values:

$$\text{Estimated Avoided Loss} = \text{Risk Events} \times P(\text{damage} \mid \text{event}) \times \text{Intervention Effectiveness} \times \text{Average Damage Cost}$$

Each term can be replaced with observed site data during a pilot. Until then, the system should report event counts, repeated-behaviour reduction, response time, and reviewed potential damage events rather than claim a monetary benefit.

Impact area	Measured mechanism
Prevention	High-risk events surfaced before the same behaviour repeats.
Quality	Faster link between handling evidence and inspection decisions.
Training	Recurring process behaviours become specific coaching topics.
Operations	Bay/process heatmaps reveal where workflow redesign may reduce risk.
Cost	Potential reduction in damage, rework, replacement, claims and investigation time; quantified only with real site data.

12. Responsible AI and Governance

Control	Implementation
Purpose limitation	Use video intelligence for damage prevention, safe handling and process improvement, not generalized employee surveillance.
No biometric identity by default	No face recognition, identity embeddings, worker ranking or automated disciplinary recommendations.
Data minimization	Store event metadata and short evidence clips rather than continuous archival copies when policy permits.
Retention	Configurable deletion windows; preserve only reviewed incidents required for operations/audit.
Access control	Role-based access and audit logs for incident viewing/export.
Human review	High-consequence or disputed incidents require supervisor review.
False-positive management	One-click disposition, threshold tuning, uncertainty display and review queue.
Explainability	Every alert exposes behaviour, evidence, risk components and timestamps.
Domain shift	Track camera/site performance; revalidate after major layout, lighting or camera changes.
LLM safety	Assistant cannot create events; structured queries retrieve stored evidence; unsupported conclusions are rejected.

13. Technical Risks and Mitigations

Risk	Mitigation
Occlusion and crowded scenes	Use track persistence, multi-frame evidence, configurable camera placement; treat severe occlusion as uncertainty.

Rare-event scarcity	Stage controlled examples, use hard negatives, augment cautiously, and avoid training/test leakage.
Camera-angle sensitivity	Calibrate zones per camera; test at least one held-out view/session.
False alerts	Multi-cue event rules, minimum durations, cooldowns, confidence gating, review feedback.
Motion blur / poor light	Resolution/fps checks, image quality warnings, camera-specific thresholds.
Unknown product weight	Use configured product class/metadata or visible size proxies; do not infer exact weight from pixels.
LLM hallucination	Tool-constrained assistant, structured event retrieval, explicit insufficient-evidence response.
Privacy concerns	Edge anonymization, no identity scoring, minimal retention, transparency, access control.
Over-scoped prototype	Lock pilot to one process and 10 behaviours; make multi-camera/digital twin future extensions.

14. Execution Plan

14.1 Six-day submission sprint (4-10 September 2026)

Sprint	Exit criterion
Day 1 - Scope and data	Lock 10 behaviours; audit organizer videos; define annotations, zones, UI wireframe and baseline data schema.
Day 2 - Perception baseline	Video ingest; detect people/products/pallet/equipment; tracking; save incident-ready trajectories.
Day 3 - Behaviour engine	Implement drop, drag, zone, stack, pallet, stepping and sequence logic; add risk score and evidence clips.
Day 4 - Productization	Dashboard, timeline, replay, alert states, grounded assistant, SOP retrieval, review feedback.

Day 5 - Evaluation	Held-out test run, per-behaviour metrics, latency, false positives, privacy checks; conduct user feedback.
Day 6 - Submission polish	Fix highest-value failures; record 3-5 representative scenarios; generate 5-6 slide deck and final architecture/impact story.

14.2 Post-submission pilot roadmap

- Weeks 1-2: expand dataset and calibrate 2-3 camera configurations.
- Weeks 3-4: improve hardest behaviours; evaluate tracking and domain shift.
- Weeks 5-6: edge deployment, retention policy, role-based access, alert tuning.
- Weeks 7-8: operational pilot, before/after repeat-event measurement, business impact review.

15. Deliverables and Judging-Criteria Alignment

Deliverable	Evidence included
Working prototype	Video ingestion, detection/tracking, 10 behaviour scenarios, risk classification, replay, dashboard, grounded explanation.
Demo evidence	3-5 representative scenarios in a concise demo, including one false-positive/human-review example.
Evaluation report	Per-behaviour metrics, false positives, latency, limitations and measured-vs-target separation.
Responsible AI note	Privacy, consent, retention, no biometric identity, review controls and assistant grounding.
Presentation deck	5-6 slides exactly mapped to problem, architecture, prototype, impact, validation and team.
Reproducibility package	Run instructions, configuration, model versions, behaviour definitions and sample event schema.

15.1 Score-maximization matrix

Criterion	Weight	Proposal evidence
Innovation & Creativity	15%	Event Graph, hybrid reasoning, uncertainty, evidence-bound assistant, privacy-first design.

Technical Execution	20%	Stable ingest/tracking, event logic, API, incident store, dashboard, replay, evaluation.
AI + Video Intelligence	20%	Detection + tracking + action/temporal reasoning + risk classification, not object detection alone.
User Experience & Feedback	10%	Supervisor-centric timeline, replay, explanations, acknowledgement and feedback; documented user iteration.
Damage Prevention & Business Impact	20%	Preventive alerting, repeat-risk reduction, auditable avoided-loss formula, process hotspots.
Presentation Quality	15%	One coherent story from activity → evidence → intervention → prevention; 3-5 high-quality demo scenarios.

16. Scope Guard: What Not to Build Before Submission

- Full digital twin / 3D warehouse visualization.
- Complex multi-camera re-identification across an entire site.
- Identity-based worker scorecards.
- Custom foundation-model training or large video-language-model fine-tuning.
- Claims of confirmed product damage from handling video alone.
- Dozens of behaviours with weak accuracy instead of 10 well-demonstrated scenarios.
- Cloud-heavy infrastructure that prevents a reliable local demo.

17. Expected Outcome

A successful HandleGuard AI prototype will demonstrate a credible transition from passive CCTV to preventive operations: the camera observes a handling sequence; the system tracks the relevant entities; temporal reasoning identifies a risky event; an interpretable risk score and evidence clip are created; a supervisor receives a proportional alert; the assistant explains the event from stored evidence; and the review outcome is captured for future improvement. The core value is not the number of detections. It is the reduction of repeated risky handling with lower investigation effort and without turning the system into identity surveillance.

18. Conclusion

The supplied challenge brief is directionally strong but incomplete as a solution proposal. HandleGuard AI makes it submission-ready by defining the product, architecture, behaviour taxonomy, risk logic, evidence model, assistant constraints, evaluation plan, responsible-AI controls, business-impact method and execution roadmap. The design is deliberately ambitious in reasoning quality but conservative in scope: one process, limited cameras and 10 behaviours. That balance gives the team a realistic path to a functioning demonstration while directly addressing every weighted judging dimension.

Appendix A - Suggested Incident Record

```
incident_id: HG-2026-00042
video_id: unloading_bay_A_0904
start_time: 00:02:14.320
end_time: 00:02:16.850
behaviour: product_drop
entities: [person_track_7, carton_track_19]
zone: unloading_transfer
risk_score: 78
risk_band: critical
model_confidence: 0.91
evidence: [pre_event_frame, impact_frame, replay_clip]
risk_components: {base:0.9, kinematics:0.8, support:0.4, duration:0.3, frequency:0.6, zone:0.8}
assistant_summary: "Potential high-risk drop observed; damage is unconfirmed. Inspect the product."
review_status: pending
reviewer_disposition: null
```

Appendix B - Literature and Technical Benchmark Notes

15. Elçi, T., & Ünlü, M. Z. (2025). An AI-based solution for warehouse safety: video surveillance system based anomaly detection in equipment-human interactions with vanilla autoencoder. *Journal of the Franklin Institute*, 362(18), 108233. DOI: 10.1016/j.jfranklin.2025.108233.
16. Önal, O., & Dandıl, E. (2024). Unsafe-Net: YOLO v4 and ConvLSTM based computer vision system for real-time detection of unsafe behaviours in workplace. *Multimedia Tools and Applications*.
17. NVIDIA Video Search and Summarization documentation (2026), RT-DETR Warehouse 2D Model v1.0.2 model/deployment note. Included as implementation evidence that warehouse-centric real-time detection pipelines are deployable; it is not a behaviour-reasoning benchmark.
18. GEG AI Video Intelligence for Warehouse Handling challenge brief (provided source, September 2026).

Appendix C - Final Submission Checklist

- ☐ 10 behaviours/scenarios are actually demonstrated, not only listed.
- ☐ At least one normal/negative scenario proves the system can stay silent.
- ☐ At least one event includes full sequence evidence rather than a single annotated frame.
- ☐ Risk and confidence are shown separately.
- ☐ Assistant answers cite incident IDs/timestamps and refuse unsupported damage claims.
- ☐ Demo includes incident replay, dashboard trends and one reviewer feedback action.
- ☐ Metrics are measured on a held-out session/video and reported per behaviour.
- ☐ No invented business savings; show formula and real inputs if available.
- ☐ Privacy slide explicitly states no face recognition / identity ranking in default design.
- ☐ Presentation contains a clear prevention story and exactly follows the 5-6 slide submission format.
- ☐ All claims in the final deck are labeled as measured result, prototype target or future work.